Using (3), we get

$$\tan\theta = \left|\frac{3-(-3)}{1-9}\right| = \frac{3}{4}$$

$$\text{Hence, } \theta = \tan^{-1}\left(\frac{3}{4}\right).$$

Example 7.15

Find the angle between the curves $y = x^2$ and $x = y^2$ at their points of intersection (0,0) and (1,1).

Solution

Let us now find the slopes of the curves.

Let m_1 be the slope of the curve $y = x^2$,

$$\text{then } m_1 = \frac{dy}{dx} = 2x.$$

Let m_2 be the slope of the curve $x = y^2$,

$$\text{then } m_2 = \frac{dy}{dx} = \frac{1}{2y}.$$

Let θ_1 and θ_2 be the angles at (0,0) and (1,1) respectively.

At $(0,0)$, we come across the indeterminate form of $0\times\infty$ in the denominator of $\tan\theta_1 = \left|\dfrac{2x-\dfrac{1}{2y}}{1+(2x)\left(\dfrac{1}{2y}\right)}\right|$ and so we follow the limiting process.

$$\tan\theta_1 = \lim_{(x,y)\to(0,0)}\left|\frac{2x-\frac{1}{2y}}{1+(2x)\left(\frac{1}{2y}\right)}\right|$$

$$= \lim_{(x,y)\to(0,0)}\left|\frac{4xy-1}{2(y+x)}\right|$$

$$= \infty$$

which gives $\theta_1 = \tan^{-1}(\infty) = \frac{\pi}{2}$.

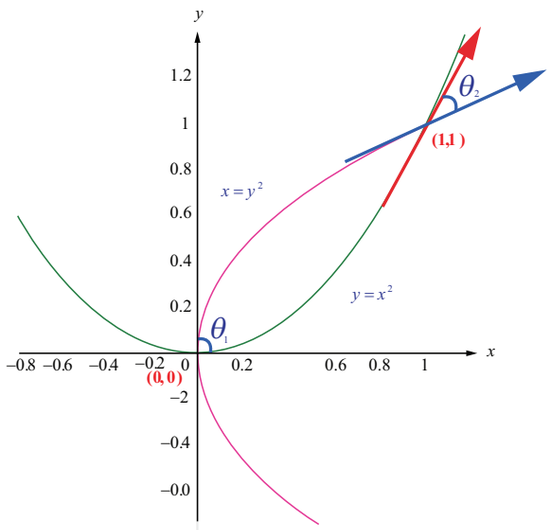


Fig.7.11

At $(1,1)$, $m_1 = 2$, $m_2 = \frac{1}{2}$

$$\tan\theta_2 = \left|\frac{2-\frac{1}{2}}{1+(2)\left(\frac{1}{2}\right)}\right|$$

$$= \frac{3}{4}$$

which gives $\theta_2 = \tan^{-1}\left(\frac{3}{4}\right)$.

12th_Maths_Vol 2_EM_CH 7_Differential Calculus.indd 12 16-12-2022 16:24:52